Unleashing UVM™

EDGE MARKER

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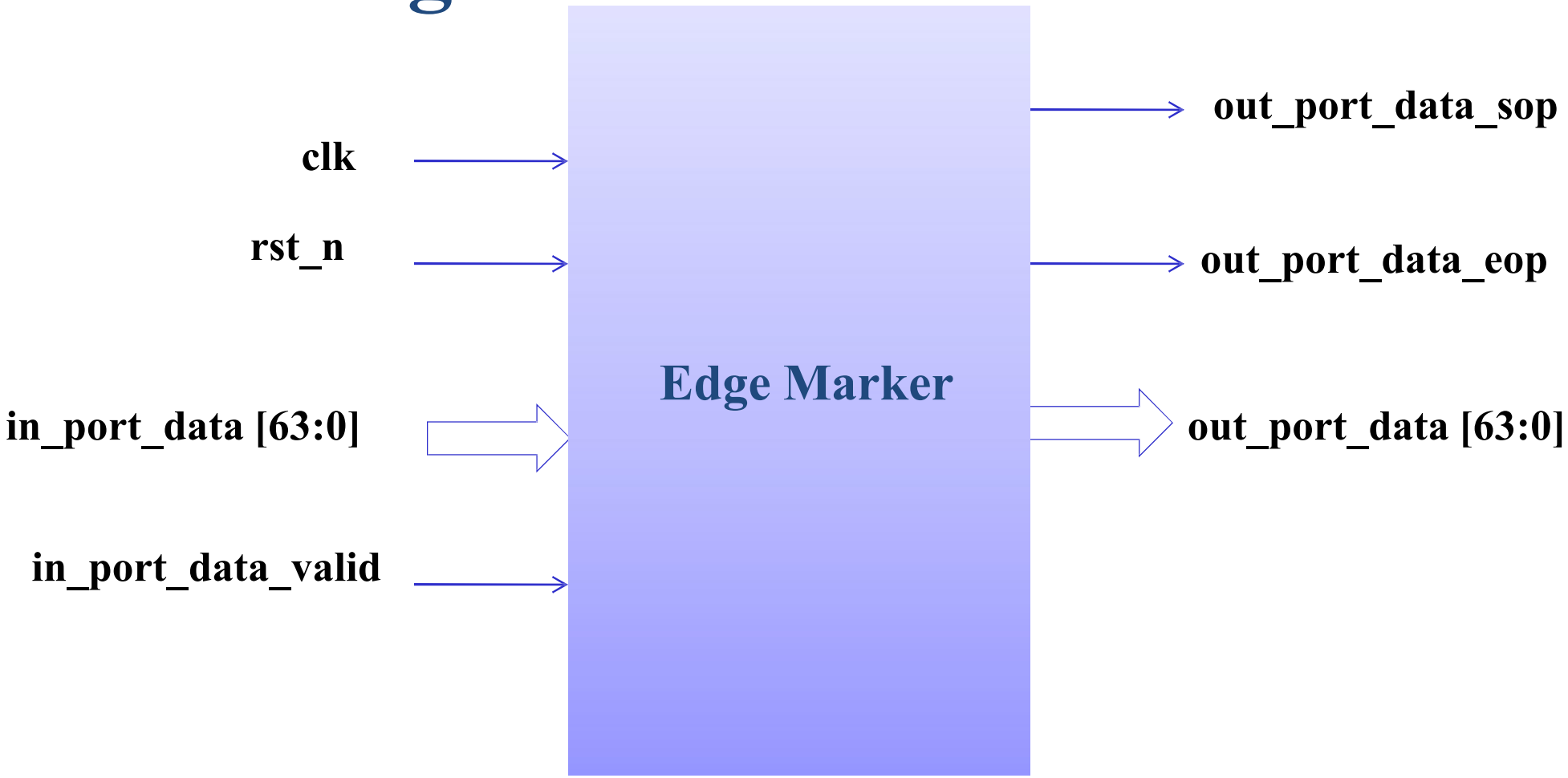
Block Diagram:



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Signal Description:

Signal	Width	Direction	Description
clk	1 bit	Input	Clock input
rst_n	1 bit	Input	Active low reset
in_port_data	64 bits	Input	Data input
in_port_data_valid	1 bit	Input	Validates input data
out_port_data	64 bits	Output	Data output
out_port_data_sop	1 bit	Output	Indicates start of packet for output data
out_port_data_eop	1 bit	Output	Indicates end of packet for output data

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Description:

Input Port :

- "in_port_data_valid" validates the data transfer .
- "in_port _data" specifies the input data to be transferred.

Output port:

- "out_port_data" fetch the data from the input port when Valid signal is high.
- "out_port_data_sop" is a pulse indicate the start of data packet transfer.
- "out_port_data_eop" is a pulse indicate the end of data packet transfer.

Note :

Variable packet size to be sent.

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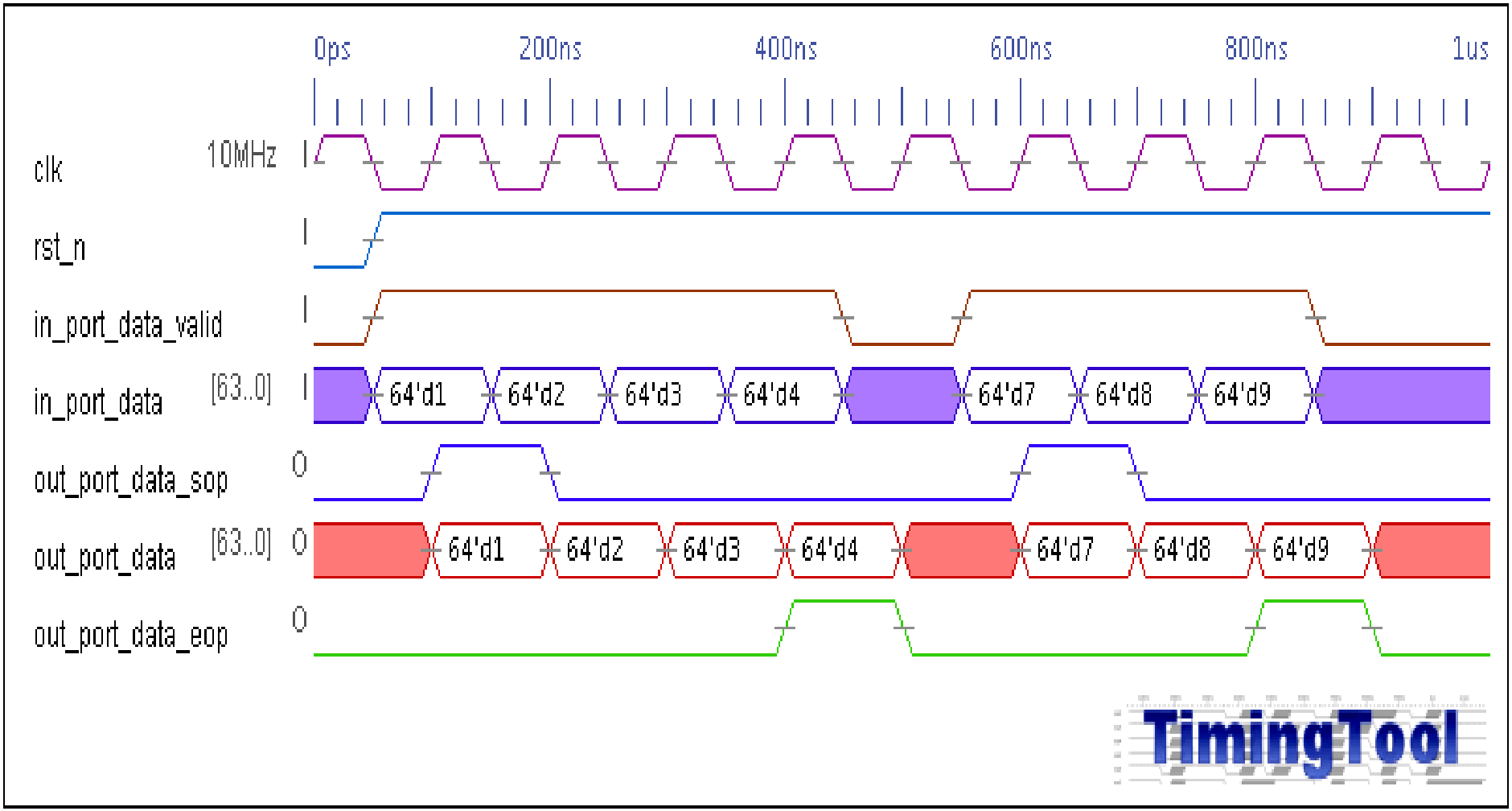
Timing_Diagram



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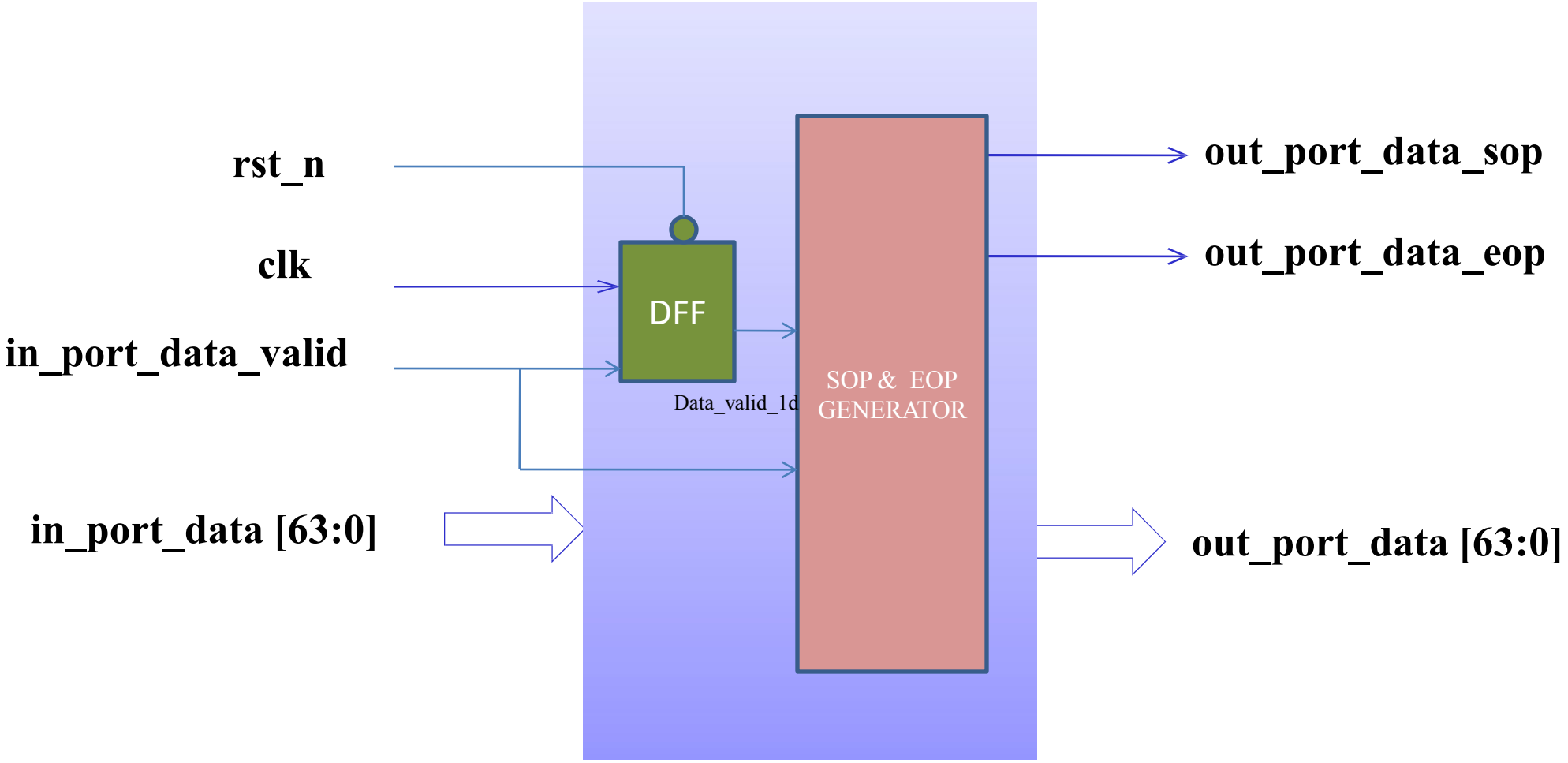
Micro Architecture:



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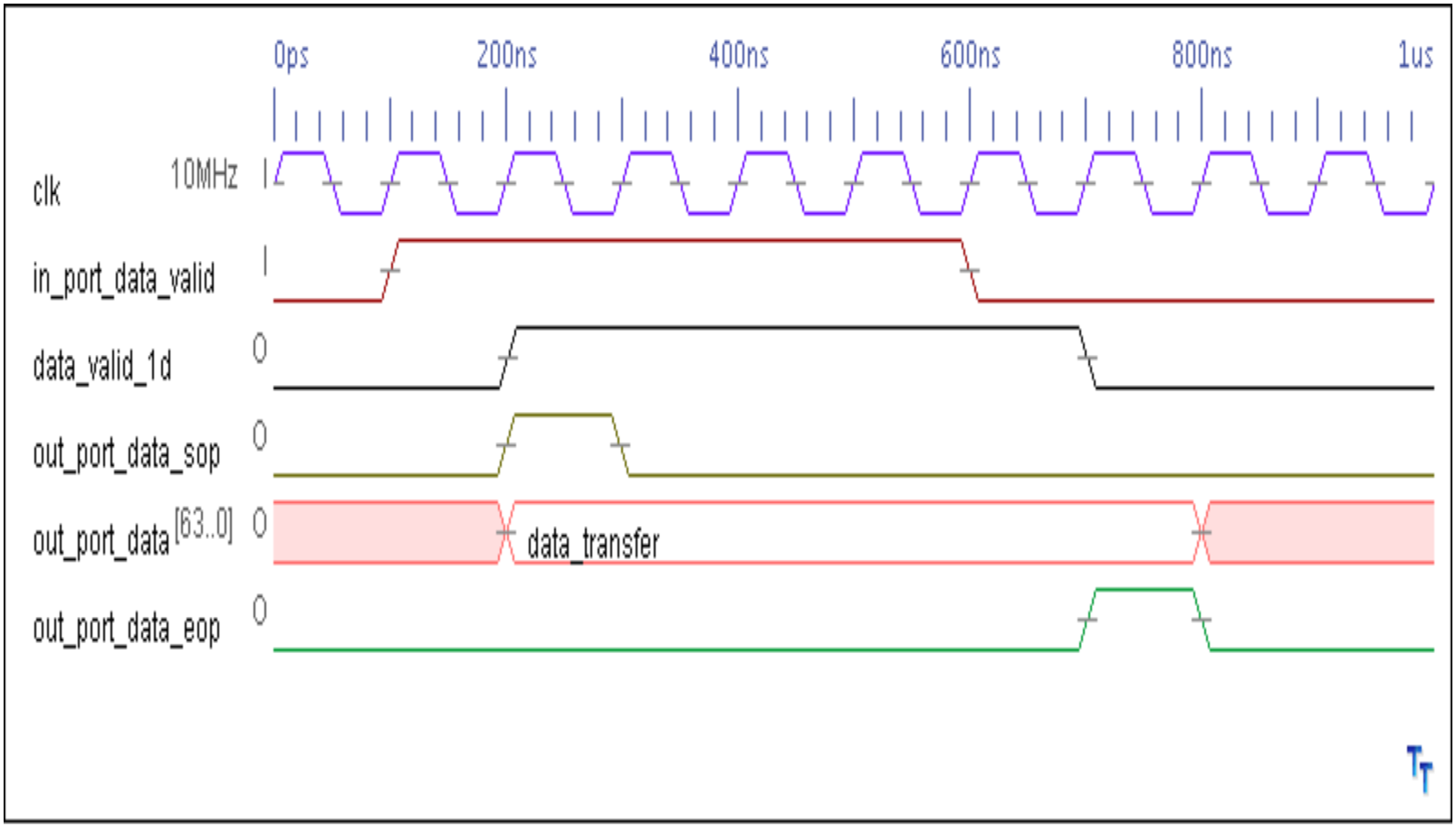
SOP & EOP Generation



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Truth table for SOP & EOP Generation

in_port_data_valid	data_valid_1d	out_port_data_sop	out_port_data_eop
0	0	0	0
0	1	0	1
1	0	1	0
1	1	0	0

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